

Linux System Performance Review

IGEL OS endpoint ITC00E0C52A6A64 | Collection: July 30, 2026, 17:53:23-17:56:44 MDT | 200-second sample

Operating system	IGEL OS 12.9.0+3
Kernel	6.18.6
CPU	4 logical CPUs
Memory	3.2 GiB RAM
Swap	1.9 GiB zram
Primary storage	7.3 GiB eMMC (/dev/mmcblk0)
Active network	Wi-Fi (wlan0), 10.0.0.18

Overall assessment: HIGH RISK / RESOURCE CONSTRAINED. During the test, the endpoint moved from moderate memory headroom to near exhaustion. Microsoft Edge renderer activity consumed most CPU and RAM, zram swap filled to 99%, and the CPU was above 90% busy in 111 of 199 one-second samples. User-facing lag, tab reloads, audio/video degradation, and application stalls are likely under this workload.

Executive Summary

Area	Status	Key evidence	Interpretation
CPU	Critical	78.0% average busy; 100% peak; >90% busy for 111/199 samples	Sustained processor saturation, not a brief spike.
Memory / Swap	Critical	Available RAM fell 1.4 GiB -> 390 MiB; swap rose 85 MiB -> 1.9 GiB	Severe memory pressure and swap thrashing.
Processes	Critical	Edge reached 92.1% CPU and 19.7% RAM in a single process; many renderers active	Browser workload is the dominant resource consumer.
Storage	Moderate	14.2% average utilization; 97.6% peak; write latency up to 194 ms	Bursty eMMC delays, amplified by swapping.
Network	Healthy	No RX/TX errors or drops; Wi-Fi carried about 67 MB RX and 11 MB TX	No evidence of link-level packet loss in this capture.
Capacity	High	Only 3.2 GiB RAM and 4 CPUs for a large multi-process browser workload	Hardware capacity is marginal for the observed session.

Detailed Findings

1. CPU Utilization

Across 199 one-second samples, total CPU busy time averaged **78.0%** and peaked at **100%**. Average user time was 47.7% and average system time was unusually high at 27.9%. The endpoint spent more than 90% busy for 111 seconds, or about 56% of the capture.

The high system CPU component is consistent with memory reclaim, compressed swap activity, browser IPC, filesystem work, and collection overhead. The monitoring tools add some load, especially pidstat, but they do not explain the sustained Edge renderer consumption or the near-full swap state.

Metric	Average	Peak
CPU busy	78.0%	100%
User CPU	47.7%	79.6%
System CPU	27.9%	72.5%
I/O wait	2.1%	33.9%
Run queue	7.8	20

2. Memory and Swap Pressure

Physical RAM is 3.2 GiB. At the beginning, 1.4 GiB was available and zram swap usage was only 85 MiB. By the end, available RAM had fallen to 390 MiB and 1.9 GiB of 1.9 GiB swap was consumed. This is the strongest evidence of the performance failure.

vmstat recorded average swap-out activity of approximately **14.4 MiB/s**, with a one-second peak near **96.2 MiB/s**. The kernel swap daemon (kswapd0) reached 12.1% CPU, confirming active memory reclaim.

Impact: compressed zram avoids slow physical swap writes, but compression/decompression consumes CPU. Once zram is full, memory allocation becomes fragile and the system may invoke the OOM killer, terminate browser processes, or become unresponsive.

3. Process Analysis

Process / group	Observed peak	Finding
Microsoft Edge	92.1% CPU; 19.7% RAM per process	Primary workload. Numerous renderer, GPU, network, extension, storage, and video-capture processes were active.
pidstat	150% CPU peak	Collection overhead; contributes load but is temporary and test-specific.
WPEWebProcess	6.1% CPU; 10.5% RAM	Significant secondary web runtime memory consumer.
mutter	4.4% CPU; 6.9% RAM	Desktop compositor; elevated memory footprint in a constrained system.
igelpkgd	4.4% CPU; 6.0% RAM	Secondary IGEL service load.
kswapd0	12.1% CPU	Kernel memory reclaim due to RAM pressure.

4. Storage I/O

The eMMC device averaged 14.2% utilization, so it was not continuously saturated. However, utilization peaked at 97.6%, read latency peaked at 112.5 ms, and write latency peaked at 194.1 ms. These bursts can cause visible pauses.

Average throughput was approximately 3.9 MiB/s read and 2.2 MiB/s write. The bursty write pattern and high merge rate are consistent with compressed filesystems, browser cache activity, and memory pressure. Storage is a contributing factor, but CPU and RAM remain the primary constraints.

Metric	Average	Peak
Reads/s	151.7	993
Writes/s	31.1	275
Read latency	2.66 ms	112.5 ms
Write latency	8.08 ms	194.1 ms
Queue depth	0.85	13.56
Device utilization	14.2%	97.6%

5. Network

The active interface was wlan0. Counters increased from 10.3 MB to 77.6 MB received and from 0.6 MB to 11.7 MB transmitted during the capture. No receive errors, transmit errors, drops, carrier errors, or collisions were reported. Ethernet enp1s0 was administratively up but had no carrier.

The socket snapshots show a large number of established HTTPS sessions owned by Microsoft Edge. This is normal for a browser with many tabs and cloud applications, but it reinforces that the browser session was large and active. The capture does not include Wi-Fi signal strength, retransmission rate, or application latency, so wireless quality cannot be fully assessed.

Prioritized Recommendations

Priority	Action	Expected benefit
1 - Immediate	Reduce Edge workload: close unused tabs/windows, disable unnecessary extensions, stop inactive web applications, and restart Edge to clear accumulated renderer state.	Fastest reduction in CPU, RAM, and socket count.
2 - Immediate	Avoid running this workload on a 4 GB-class endpoint. Move to at least 8 GiB RAM; 16 GiB is preferable for many browser tabs, video meetings, or multiple web runtimes.	Prevents zram exhaustion and reclaim storms.
3 - High	Review Edge policies and applications for background tabs, startup boost, preloading, sleeping-tab exclusions, video capture, and extension behavior.	Reduces persistent browser process growth.
4 - High	Reproduce with a lighter collection profile. Run mpstat/vmstat/iostat at 2-5 second intervals and omit per-thread pidstat unless needed.	Separates workload behavior from measurement overhead.
5 - Medium	Collect per-process PSS/RSS using smem or /proc/*/smaps_rollup, plus Edge task-manager data, before and after opening the business application.	Identifies the specific tab, extension, or renderer causing growth.

Priority	Action	Expected benefit
6 - Medium	Capture zramctl, /proc/meminfo, pressure stall information (/proc/pressure/*), dmesg OOM messages, and Wi-Fi station statistics.	Confirms memory stalls, compression ratio, OOM risk, and wireless health.
7 - Medium	If eMMC latency remains high after memory pressure is resolved, inspect filesystem free space, Btrfs allocation, browser cache, and device health.	Determines whether storage is independently degrading performance.

Conclusion

This endpoint is under-provisioned for the observed browser workload. The dominant failure mode is the combination of Edge CPU demand and insufficient physical memory. The resulting zram saturation and kernel reclaim increase system CPU and create intermittent storage latency. Network counters are clean, and the eMMC device is not continuously saturated. Reducing the browser workload or increasing RAM should be addressed before investigating storage or network as primary causes.

Confidence: High for CPU, memory, process, and block-device conclusions. Moderate for network quality because wireless signal and retransmission statistics were not collected. The capture represents one 200-second interval and should be repeated after remediation for comparison.